

TÉCNICO SUPERIOR UNIVERSITARIO EN TECNOLOGÍAS DE LA INFORMACIÓN ÁREA DESARROLLO DE SOFTWARE MULTIPLATAFORMA

NOSQL

Quinto A

ENTREGABLE
PYTHON VS MONGO ATLAS

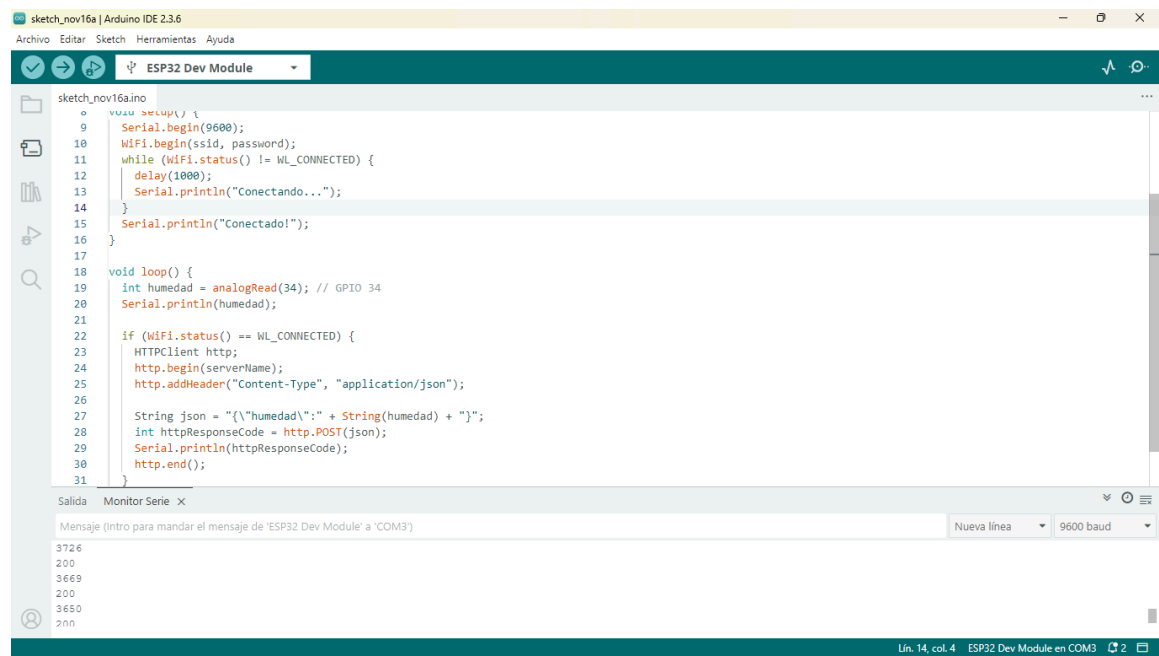
UNIVERSIDAD TECNOLÓGICA DE JALISCO
PRESENTA:

Luis Rodolfo Silva Castellanos

PROFESOR:
CARLOS IVAN

GUADALAJARA, JALISCO, 16 DE NOVIEMBRE DE 2025

CÓDIGO ARDUINO



```

1  sketch_nov16a.ino | Arduino IDE 2.3.6
2  Archivo  Editar  Sketch  Herramientas  Ayuda
3  ESP32 Dev Module
4
5  sketch_nov16a.ino
6
7  void setup() {
8    Serial.begin(9600);
9    WiFi.begin(ssid, password);
10   while (WiFi.status() != WL_CONNECTED) {
11     delay(1000);
12     Serial.println("Conectando...");
13   }
14   Serial.println("Conectado!");
15 }
16
17 void loop() {
18   int humedad = analogRead(34); // GPIO 34
19   Serial.println(humedad);
20
21   if (WiFi.status() == WL_CONNECTED) {
22     HTTPClient http;
23     http.begin(serverName);
24     http.addHeader("Content-Type", "application/json");
25
26     String json = "{\"humedad\": " + String(humedad) + "}";
27     int httpResponseCode = http.POST(json);
28     Serial.println(httpResponseCode);
29     http.end();
30   }
31 }

```

Salida Monitor Serie x

Mensaje (Intro para mandar el mensaje de 'ESP32 Dev Module' a 'COM3')

3726
200
3669
200
3650
200

Un. 14, col. 4 ESP32 Dev Module en COM3

```

#include <WiFi.h>
#include <HTTPClient.h>

const char* ssid = "Laimposible2.4";
const char* password = "H2Oco2HJems";
const char* serverName = "http://192.168.1.87:5000/humedad";

void setup() {
  Serial.begin(9600);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Conectando...");
  }
  Serial.println("Conectado!");
}

```

```

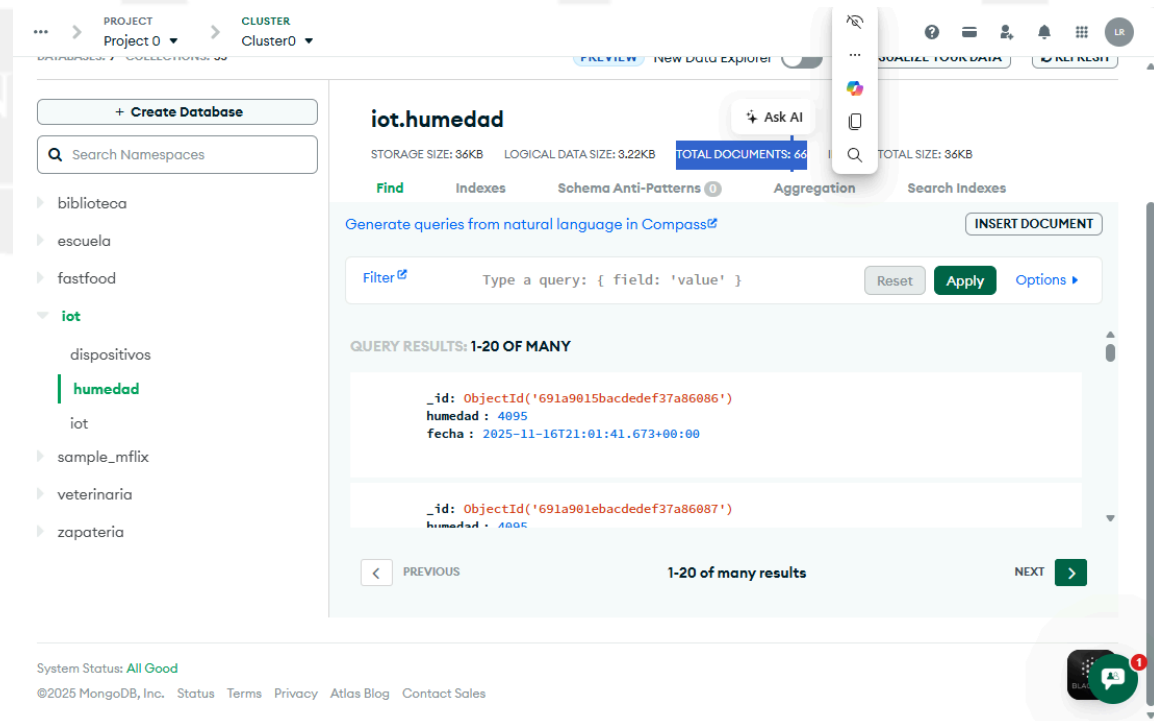
void loop() {
  int humedad = analogRead(34); // GPIO 34
  Serial.println(humedad);

  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    http.begin(serverName);
    http.addHeader("Content-Type", "application/json");

    String json = "{\"humedad\":\"" + String(humedad) + "\"}";
    int httpResponseCode = http.POST(json);
    Serial.println(httpResponseCode);
    http.end();
  }

  delay(5000);
}

```



The screenshot shows the MongoDB Atlas web interface. On the left, there's a sidebar with a search bar and a list of namespaces including 'biblioteca', 'escuela', 'fastfood', 'iot', 'sample_mflix', 'veterinaria', and 'zapateria'. The 'iot' namespace is expanded, showing 'dispositivos', 'humedad', and 'iot'. The 'humedad' collection is selected.

The main panel displays the 'iot.humedad' collection details:

- STORAGE SIZE: 36KB
- LOGICAL DATA SIZE: 3.22KB
- TOTAL DOCUMENTS: 66
- TOTAL SIZE: 36KB

Below the details, there's a 'Filter' section with a query input field containing '{ field: 'value' }'. Below that, the 'QUERY RESULTS: 1-20 OF MANY' are shown. The results are displayed in a table-like format with two rows of data:

_id	humedad	fecha
ObjectId('691a9015bacedef37a86086')	4895	2025-11-16T21:01:41.673+00:00
ObjectId('691a901ebacedef37a86087')	4895	

At the bottom of the interface, there's a 'System Status: All Good' message and a footer with copyright information for MongoDB, Inc. and links to Status, Terms, Privacy, Atlas Blog, and Contact Sales.

CODIGO PARA HACER EL POST A MONGO BD ATLAS

```
from flask import Flask, request
from pymongo import MongoClient
from datetime import datetime

app = Flask(__name__)
cliente =
MongoClient("mongodb+srv://2124200328_db_user:goTe8610@cluster0.hr2u
uz8.mongodb.net/?appName=Cluster0")
db = cliente["iot"]
coleccion = db["humedad"]

@app.route('/humedad', methods=['POST'])
def recibir():
    data = request.get_json()
    data["fecha"] = datetime.now()
    coleccion.insert_one(data)
    return {"status": "ok"}

app.run(host='192.168.1.87', port=5000)
```

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CÓDIGO PARA IMPORTAR CSV Y GRAFICAR

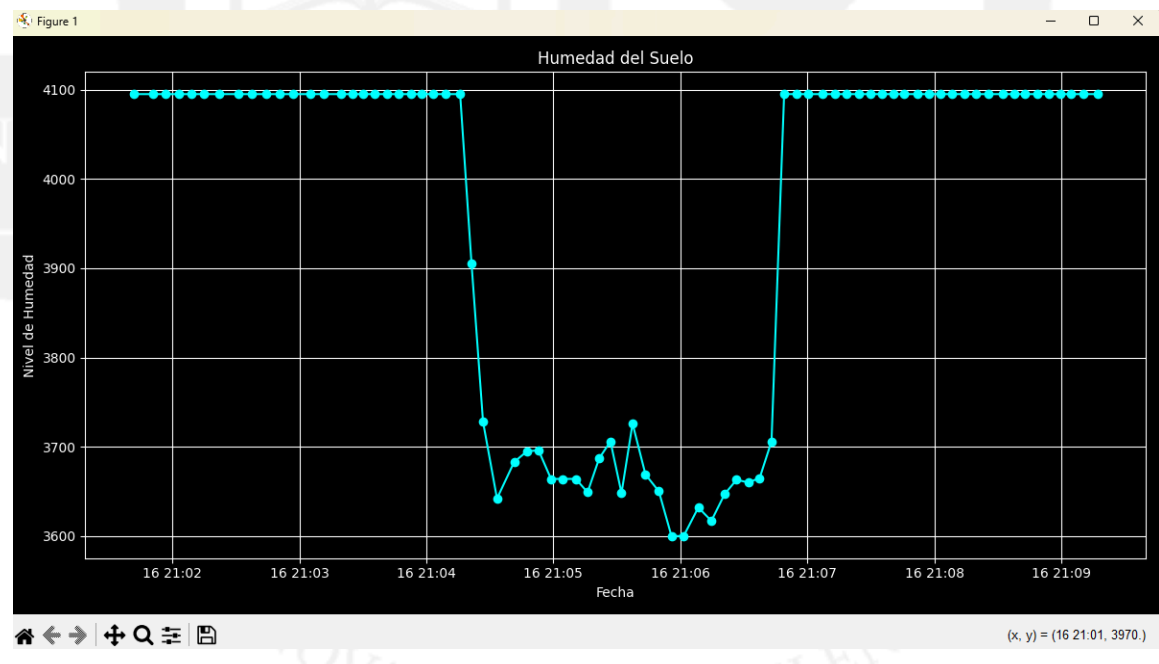
```
import pandas as pd
from pymongo import MongoClient
import matplotlib.pyplot as plt

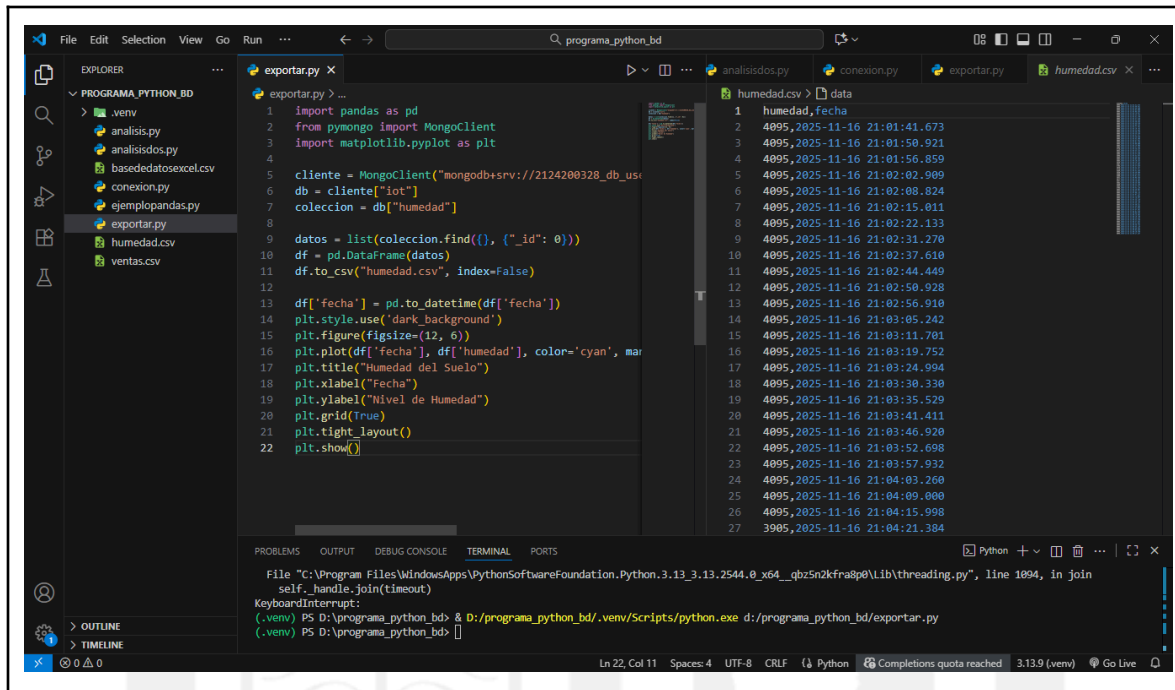
cliente =
MongoClient("mongodb+srv://2124200328_db_user:goTe8610@cluster0.hr2u
uz8.mongodb.net/?appName=Cluster0")
db = cliente["iot"]
```

```
coleccion = db["humedad"]

datos = list(coleccion.find({}, {"_id": 0}))
df = pd.DataFrame(datos)
df.to_csv("humedad.csv", index=False)

df['fecha'] = pd.to_datetime(df['fecha'])
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))
plt.plot(df['fecha'], df['humedad'], color='cyan', marker='o')
plt.title("Humedad del Suelo")
plt.xlabel("Fecha")
plt.ylabel("Nivel de Humedad")
plt.grid(True)
plt.tight_layout()
plt.show()
```





```

1 import pandas as pd
2 from pymongo import MongoClient
3 import matplotlib.pyplot as plt
4
5 cliente = MongoClient("mongodb+srv://2124200328_db_usi
6 db = cliente["iot"]
7 coleccion = db["humedad"]
8
9 datos = list(coleccion.find({}, {"_id": 0}))
10 df = pd.DataFrame(datos)
11 df.to_csv("humedad.csv", index=False)
12
13 df["fecha"] = pd.to_datetime(df["fecha"])
14 plt.style.use('dark_background')
15 plt.figure(figsize=(12, 6))
16 plt.plot(df["fecha"], df["humedad"], color='cyan', mar
17 plt.title("Humedad del Suelo")
18 plt.xlabel("Fecha")
19 plt.ylabel("Nivel de Humedad")
20 plt.grid(True)
21 plt.tight_layout()
22 plt.show()

```

humedad.csv > data

```

1 humedad,fecha
2 4095,2025-11-16 21:01:41.673
3 4095,2025-11-16 21:01:50.921
4 4095,2025-11-16 21:01:56.859
5 4095,2025-11-16 21:02:02.909
6 4095,2025-11-16 21:02:08.824
7 4095,2025-11-16 21:02:15.011
8 4095,2025-11-16 21:02:22.133
9 4095,2025-11-16 21:02:31.270
10 4095,2025-11-16 21:02:37.610
11 4095,2025-11-16 21:02:44.449
12 4095,2025-11-16 21:02:50.928
13 4095,2025-11-16 21:02:56.910
14 4095,2025-11-16 21:03:05.242
15 4095,2025-11-16 21:03:11.781
16 4095,2025-11-16 21:03:19.752
17 4095,2025-11-16 21:03:24.904
18 4095,2025-11-16 21:03:30.330
19 4095,2025-11-16 21:03:35.529
20 4095,2025-11-16 21:03:41.411
21 4095,2025-11-16 21:03:46.920
22 4095,2025-11-16 21:03:52.698
23 4095,2025-11-16 21:03:57.932
24 4095,2025-11-16 21:04:03.260
25 4095,2025-11-16 21:04:09.000
26 4095,2025-11-16 21:04:15.998
27 3905,2025-11-16 21:04:21.384

```

File "C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.13.3.13.2544.0_x64__qbz5n2kfra8p0\Lib\threading.py", line 1094, in join
self._handle.join(timeout)
KeyboardInterrupt:
(.venv) PS D:\programa_python_bd> & D:/programa_python_bd/.venv/Scripts/python.exe d:/programa_python_bd/exportar.py
(.venv) PS D:\programa_python_bd> []

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INNOVACIÓN Y EXCELENCIA